Neffke et al. (2017): Inter-industry labor flows

Research problem: The paper is based on the observation that regions tend to grow new industries that are close to the ones they already have, but we don't know how to measure how "close" two industries are.

What earlier relatedness measures missed: Earlier studies tried to use co-exports, co-location and input-output links, but these don't show clearly whether two industries need the same worker skills. They can also be misleading, such as industries simply being in the same area, plus they missed which industries swap workers, since they assumed all industries are equally far apart, so they saw many switches and simply concluded that skills are not tied to an industry.

Question: How industry-specific is human capital and which industries have similar human capital requirements?

Contribution: The authors show that moves are concentrated in few industry pairs, stable over time and similar across worker types, their skill-relatedness measure predicts local industry growth and entry better than co-location and input-output measures. To facilitate future research, the authors made detailed inter industry relatedness matrices online available.

Related Papers:

- **O'Clery & Kinsella (2022)**: The paper's main extension is the method, and the data form the UK. They use job moves between industries to make a map of industries. They then add a new method that finds groups of industries that swap many workers, which they call "skill basins".
- **Elekes et al. (2023)**: The main extension is a new outcome on worker pay mobility and the data from Sweden. They ask whether low-wage workers move to better-paid jobs when their region has more skill-related high-income jobs, they use data on individual workers instead of industry totals, and they find that the answer is yes.

Gap: Previous work has shown that the skill-relatedness map is stable over time and that it predicts which industries grow and which workers move up. But we do not know whether the map stays the same over a longer time since Neffke et al. looked only at 1999–2008, and the two later papers assume the map stays fixed. This matters because regions and training programmes use these maps to plan, so if industries change and the links between industries change, old maps give wrong advice.

Research question:

How much did the skill-relatedness links between Hungarian industries change between 2003–2009 and 2011–2017?

Data that could answer research question: Admin3 panel from the KRTK Databank (It has monthly data on jobs with firm information linked to each worker)

Who could use it: Hungarian policymakers and regional development agencies.